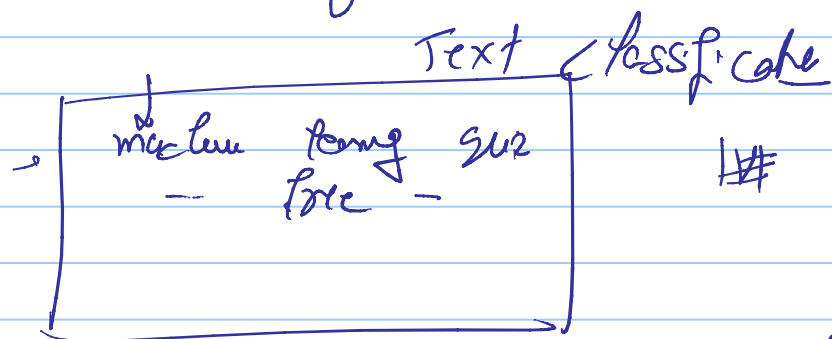


COL 774 / COL 734
Sep 9, 2025

Naive Bayes:



$\{w_1, \dots, w_{|V|}\}$

$\{x_i^u, y_i^u\}_{i=1}^m$

$y_i^u \in \{0, 1\}$

Simplest Case:

$x_1^u, \dots, x_n^u$

$x_j^u = \begin{cases} 1 \\ 0 \end{cases}$

if w_j occurs in doc(i)

$p(y|w)$ $p(y|y)$ $\mathcal{D}, \mathcal{D}'$

$y \sim \text{Multinomial}(\mathcal{D})$

$y \sim \text{Multinomial}(\mathcal{D})$

$\mathcal{D}_j | y$

$\sum_{j=1}^J \mathcal{D}_j | y = 1$

$\hookrightarrow n = |V|$ # of attributes

(1) $y \sim \text{Bernoulli}(\phi) :-$ class prob. $\rightarrow$

$p(y)$

$\phi =$

$p(x/y)$

Naive Bayes
assumption
 $x_1 \perp x_2 \perp y$

(2) $x_1, x_2, \dots, x_n | y \sim \text{Bernoulli}(\phi_j | y)$

Bernoulli count model

MLE

Maximum likelihood estimate

$$\phi = \frac{\sum_{i=1}^m 1 \{y^{(i)} = 1\}}{m}$$

$$\frac{\sum_{i=1}^m 1 \{y^{(i)} = 1\} \sum_{j=1}^n x_j^{(i)}}{\sum_{i=1}^m 1 \{y^{(i)} = 1\}}$$

Bag of words model

$$q | y = 1 =$$

$$\frac{\sum_{i=1}^m 1 \{y^{(i)} = 1\} \sum_{j=1}^n x_j^{(i)}}{\sum_{i=1}^m 1 \{y^{(i)} = 1\}}$$

Smoothing $\hookrightarrow q | y = 0$
Laplace

param $(2 \times n + 1)$

λ Inference.

~~$P(x)$~~

$P(y|x)$

λ

$P(x/y) P(y)$

$$\rightarrow \arg\max_y P(y|x) = \arg\max_y P(x/y) P(y)$$

$$\downarrow$$

$$\prod P(x_i/y_i)$$

$$\downarrow$$

- ① does not need sequence x
 ② does not need frequency ✓

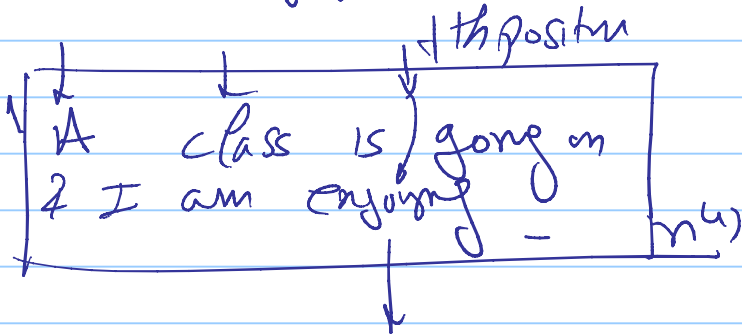
Model :- Bag of words model

$y \sim \text{Bernoulli}(p)$

$y_i/y \sim \text{Poisson}(\lambda_i/y)$

Try it out?

$\Rightarrow$



$\Rightarrow d(u)$

$y^{(u)} \sim \text{Bernoulli}(p)$
 $u \in 1$

$$w \text{ Multinoulli}(O_{j|y})$$

$$\sum_{c=1}^{|V|} O_{c|y} = 1$$

$$\#_{\mathcal{D}} \quad x_j^w \text{ w Multinoulli}(O_{j|y})$$

$\downarrow$
 $\{1 - n\}$

eg:- $\mathcal{D}_1, \mathcal{D}'$

$$O_{j|y} \equiv O_{j'|y}$$

$\hookrightarrow \# \text{ params}$

(Param for tying)
assumption

$$O(|V| \times C) \# \text{ classes}$$

Multinoulli event model

Bag of words model

$$\phi = \frac{\sum_i 1 \sum_j y_{ij} = 13}{m}$$

$\rightarrow \# \text{ words on its count}$

$$\#:- O_{c|y=1} =$$

Laplace smoothing $\rightarrow$

$$\left[\sum_{i=1}^m \right]$$

$$\frac{1}{m} \sum_{j=1}^n \left[\sum_{c=1}^{|V|} 1 \sum_{j=1}^n x_{cj}^w = c \right] + 1$$

$\# \text{ of times } c^{\text{th}} \text{ word occurs}$
 $n_j^w \text{ doc}(w)$

$$\hookrightarrow \sum P(y|x) \propto \alpha$$

$$\prod_{j=1}^n P(x_j|y) P(y)$$

$$\left[\sum_{i=1}^m \mathbb{1}_{\{y^i = z\}} n^{(i)} \right] + |V|$$

↓ # of words in 1st column

Take a big word
counting